Alexander Gimourginas

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Education

Seton Hall University May 2027
Bachelor of Science in Computer Science *South Orange, NJ*

- **Awards**: Honors College, Dean's List, MLK Leader Scholarship, University Scholarship
- **Relevant Coursework**: Python & Java (Object-Based), Scheme, Linear Algebra, Discrete Mathematics, Operating Systems, Programming Languages, Computer Networking, Data Mining, Big Data Analytics, Automata Theory

Experience

Pirate Shield - SOC Lab Platform Development Summer 2026 - Present
Software Engineer *Seton Hall University*

- Build the validation environment for PirateShield, a security platform for New Jersey K-12 schools: a simulated school with staff accounts across six roles (teacher, counselor, nurse, front office, administrator, IT) and synthetic student records.
- Engineer automated scenario scripts that simulate a normal workweek and four classes of insider misuse (off-hours, cross-role, and bulk record access; stolen-credential sign-ins), each with a predefined expected outcome to score true and false alerts.
- Prepare a Dec. 2026 leadership demo in support of a pilot funding request, with acceptance criteria of 90%+ of misuse flagged and at most 1 false alert per 100 normal sessions; contribute to the district pilot plan.

Pirate Shield - Endpoint Risk Model Research Fall 2025 - Summer 2026
Security and ML Research Assistant *Seton Hall University*

- **First-author paper accepted at ACM RACS '26** (International Conference on Resilience, AI and Cyber-Systems), Fukuoka, Japan, Nov. 17-20, 2026: *A Three-Layer Composite Risk Model for Endpoint Security Monitoring in K-12 Schools.* doi:10.1145/3848591.3856865
- Designed and built a 3-layer endpoint risk model in TypeScript and Python for a K-12 cybersecurity platform: weighted rule-based scoring, an unsupervised autoencoder trained only on normal system-call activity, and Hawkes-inspired temporal chain scoring, fused into a bounded 0-100 score with five severity tiers and under 3 ms scoring time per event.
- Evaluated the autoencoder on ADFA-LD (10-fold cross-validation, mean AUC 0.76) and ran a layer ablation on multi-step attack scenarios; temporal correlation added the most signal, with no benign event reaching medium severity.

Projects

Dungeon Shop - Roblox Dungeon-Extraction and Shop Game Sep. 2026 - Present
Luau, Roblox, Rojo, ProfileStore, Git

- Architected a server-authoritative, data-driven game: enemies, items, depths, and events are config modules, so new content ships without touching gameplay logic.
- Developed procedural dungeon generation, combat and enemy AI, loot and inventory, and a shop with customer selling loop, with persistent player data via ProfileStore.
- Set up a pre-commit gate (StyLua, selene, luau-lsp) and test modules for the generator, combat, and run manager.

LoanLens - Debt Payoff Strategizer Web App Mar. 2026
Next.js, TypeScript, Tailwind CSS, Supabase, Anthropic API

- Led a 5-person team to build and deploy a full-stack debt payoff app in 24 hours, placing 4th at the SHU 2026 Hackathon.
- Integrated the Anthropic API via Next.js API routes to deliver personalized repayment strategies from user loan data.
- Built the onboarding flow, payoff dashboard, and what-if simulator with real-time amortization and data visualizations.
- Persisted loan data in Supabase with UUID-based shareable dashboard links; added email notifications via Resend.

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, Luau, Scheme, SQL, HTML, Racket

Tools: Next.js, React.js, Tailwind CSS, Supabase, Anthropic API, SQLite3, Git, GitFlow, OpenAI API, Tkinter, Swing, LaTeX

Security & ML: Anomaly Detection, Autoencoders, Host-Based Intrusion Detection, Event Correlation, Risk Scoring

Data & Cloud: AWS (IAM, EC2, EBS, VPC), scikit-learn, pandas

Leadership Positions

Executive Board Member, ACM SIGPLAN - Seton Hall Chapter Sep. 2024 - Jan. 2025

Organized and led bi-weekly programming language workshops and coordinated guest speakers.

MLK Leadership Program, Seton Hall Jun. 2023 - Present

Completed 50+ hours of leadership development and community service, including mentorship and outreach; led a high-turnout blood drive and fundraising for a recreational facility.